



Track A — Data Cleaning, Quality Issues, and Spend Classification

1. Objective

The cleaning and classification process was designed to satisfy two requirements:

1. **Flag data-quality issues and document how each issue was handled.**
2. **Clean and classify every purchased part into a small, sensible set of categories, with explicit reasoning for borderline cases.**

The approach therefore separates:

- **Data-quality remediation:** missing values, typos, inconsistent grade labels, and other structural issues.
- **Business classification:** determining what each part actually represents from its description and assigning it to a consistent category.

The original fields are retained wherever practical so that every correction remains traceable.

2. Data-Quality Issues Found and Treatment

Issue found	Evidence / check	Treatment	Reason
3 missing values in category	Missing-value audit of the original category field	Corrected / imputed	The part descriptions provided enough semantic

Issue found	Evidence / check	Treatment	Reason
			information to classify these parts without using a frequency-based imputation such as the mode
Farsteners typo	Category frequency and label inspection	Corrected to Fasteners	Deterministic spelling correction; otherwise the same business category would appear as two categories
Inconsistent gradetype values	Inspection of grade/material labels	Standardized / corrected	Equivalent grade representations should map to one consistent value before material-price calculations
Category assigned inconsistently with description	Cross-check between <code>category</code> , <code>description_bin</code> , and <code>part_description</code>	Corrected where evidence was clear	The original category is a historical label, so it was not treated as ground truth when it contradicted the actual part description
Material type provides little category differentiation	<code>type_of_material</code> reviewed across records	Used as a consistency check, not a classification	Material was consistently steel, so it did not distinguish the

Issue found	Evidence / check	Treatment	Reason
		driver	physical/component category
Rare categories	Category frequency review	Reviewed individually; not automatically removed	Low frequency does not by itself mean a category is incorrect
Numerical fields used for costing	Review of weight, volume, unit price and derived price-per-kg fields	Validated before cost modeling	Prevents unit or value inconsistencies from propagating into the should-cost calculations

Treatment Philosophy

The general rule was:

- **Correct** deterministic issues such as typos and inconsistent representations.
 - **Impute** missing categorical values only when the part description provides sufficient evidence.
 - **Reclassify** categories when the historical label conflicts with clear semantic evidence.
 - **Retain and review** ambiguous cases instead of forcing unsupported classifications.
 - **Flag for confirmation** when the available data cannot support a confident decision.
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3. Category Classification Methodology

3.1 Primary source of classification

The `part_description` was treated as the primary source for understanding what the purchased part actually is.

Examples of useful component terms included:

- Bracket
- Mount
- Panel
- Plate
- Beam
- Nut
- Rail
- Pillar

The description often provides stronger evidence of the physical component type than the historical category field.

3.2 Description-bin construction

A `description_bin` column was created from `part_description` using regex-based extraction.

The broad bins were:

Panel
Bracket
Plate
Mount
Beam
Nut
Rail
Pillar

The purpose of this field was **not to become the final business taxonomy automatically**. It was created as an intermediate, interpretable feature for:

- identifying the physical/component type,
- imputing the 3 missing categories,
- detecting inconsistencies between description and historical category,
- supporting manual review of borderline records.

4. Handling the 3 Missing Categories

The 3 missing category values were handled using a **description-driven imputation process**.

For each missing category:

1. Inspect the `part_description`.
2. Extract the relevant component type through `description_bin`.
3. Compare the component description with the surrounding cleaned category structure.
4. Assign the category that best represents the actual part.

This was preferred over:

Missing category → most frequent category

because the mode would ignore the specific identity of the part.

The decision rule was:

Use the part's observable physical/component description rather than the overall frequency of historical labels.

5. Detecting Incorrect Existing Categories

After constructing `description_bin`, the records were cross-examined to identify cases where:

```
part_description
    ≠
existing category
```

Examples of the logic include:

- A description containing an explicit **Bracket** indication should not remain in an unrelated category without evidence.
- A description clearly identifying a **Panel** should not be categorized as a generic or unrelated structural type without justification.
- A description identifying a **Nut/Fastener** should be treated as hardware rather than as a broad formed-metal category.

The original category was therefore treated as **evidence**, not as unquestionable truth.

6. Borderline-Case Decision Rules

Rule 1 — Prefer explicit component identity

If the description clearly identifies the part type, use that information first.

Example:

Engine Mount Bracket
→ Bracket

rather than relying on an inconsistent historical label.

Rule 2 — Use the functional/component noun rather than the application location

Words such as:

Door
Hood
Seat
Floor
Engine
Battery

describe where the component is used.

Words such as:

Bracket
Panel
Plate
Beam
Rail
Nut

more directly describe what the component is.

Therefore, the component type was prioritized when defining the category.

Rule 3 — Use `type_of_material` as a validation field

The material field was reviewed together with the description.

Because the dataset was consistently identified as **steel**, material type did not provide enough information to distinguish the component categories.

Therefore:

Description → primary classification evidence
Material → consistency check

Rule 4 — Do not create unnecessary categories for every noun

The objective was not to create one category for every phrase appearing in a description.

For example, these can all remain in the broader structural grouping:

Engine Mount Bracket
Chassis Mount Bracket
Steering Column Bracket

because their common component form is a bracket.

Rule 5 — Preserve ambiguity

If a description does not contain enough evidence to assign a category confidently, it should be flagged for client confirmation rather than being assigned arbitrarily.

7. Final Cleaned Category Structure

After correcting inconsistencies, the detailed categories were consolidated into a smaller manufacturing-oriented taxonomy.

Cleaned categories	Condensed category	Reason for grouping
Brackets & Mounts	Structural Formed Components	Similar structural/forming behavior and meaningful conversion effort
Reinforcements	Structural Formed Components	Structural components with forming/reinforcement requirements
Rail	Structural Formed Components	Structural formed component with similar broad manufacturing considerations
Body Panels	Body Panels	Kept separate because panel forming can have materially different characteristics
Fasteners	Hardware	Hardware/fastening component
Nut	Hardware	Same broad hardware family, while detailed category is retained

Cleaned categories	Condensed category	Reason for grouping
Plate	Plates	Generally simpler flat/formed sheet-metal component

Final condensed groups:

Structural Formed Components
Body Panels
Hardware
Plates

The detailed cleaned category is retained alongside the condensed category for traceability.

8. Why the Condensed Categories Are Used

The condensed categories are used for the **should-cost model**, not to erase useful information.

They provide a manageable set of manufacturing groups for assigning assumptions such as:

- relative conversion complexity,
- cycle-time ranges,
- manufacturing treatment.

This prevents the should-cost model from becoming overly fragmented.

9. Additional Cleaning and Validation Steps

The following checks were also performed before cost modeling:

Missing-value audit

Reviewed missing values across the dataset rather than assuming that only category values required attention.

Category-frequency audit

Examined category frequencies to identify:

- rare categories,
- duplicated spellings,
- inconsistent labels,
- potentially broad or suspicious classifications.

Grade normalization

Standardized inconsistent `gradetype` representations before using grade information in material-price calculations.

Numerical consistency

Reviewed:

- `unit_weight_kg`
- `annual_volume`
- `unit_price_local`
- `spend`
- `per_kg_price`

and the associated derived material-cost fields before using them in the should-cost model.

Traceability

The original category information was not discarded. Cleaned and condensed fields were created separately so that changes can be audited.

10. Final Data-Cleaning Workflow

```
Raw Dataset
  ↓
Missing-value audit
  ↓
Category-frequency and label audit
  ↓
Correct deterministic typo
  ↓
Standardize gradetype inconsistencies
  ↓
Create description_bin from part_description
  ↓
Impute 3 missing categories from description evidence
  ↓
Cross-examine existing categories against descriptions
  ↓
Correct clearly misplaced categories
  ↓
Use material type as a consistency check
  ↓
Review rare / borderline cases
  ↓
Create condensed manufacturing categories
  ↓
Validate numerical inputs for cost modeling
  ↓
Clean dataset ready for should-cost analysis
```

11. Final Outcome

The cleaning process was designed to produce a dataset in which:

- **Every part has a category.**
- Category labels are standardized.
- Obvious historical labeling errors are corrected.
- Material-grade inconsistencies are normalized.
- Category assignments are supported by the actual part description.
- Borderline classifications are handled explicitly rather than hidden behind an automated output.
- Detailed categories are retained for traceability.
- Condensed categories provide a small, sensible taxonomy for spend and should-cost analysis.